

PALETTE: Partition-based Affine Local-intEnsity Transform for Texture-prEserving augmentation

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Abstract

001 Segmentation models for medical images are trained on
002 one acquisition setting and deployed on others, and label-
003 generative domain randomization (SynthSeg) addresses
004 this by resampling intensities from a per-label Gaussian
005 mixture over a dense anatomical label map – an ap-
006 proach that is both label-driven (it requires exactly the
007 expensive annotation the method is trying to reduce re-
008 liance on) and texture-destroying (it replaces real gradi-
009 ents and partial-volume detail with piecewise noise). We
010 introduce PALETTE (Partition-based Affine Local-intEnsity
011 Transform for Texture-prEserving augmentation), a single-
012 source, label-free augmentation that partitions the fore-
013 ground of a scan into intensity classes with unsupervised
014 k-means, further subdivides each class into spatial sub-
015 regions with a Voronoi tessellation, and applies a signed
016 random affine remap $y = \mu + \alpha(x - \bar{x})$, $\alpha \in \pm U(0.5, 2.0)$,
017 to the real intensities within each sub-region. Because the
018 transform remaps existing intensities rather than replacing
019 them with sampled noise, edges, gradients, and partial-
020 volume detail are preserved within each sub-region; only
021 the region-to-region contrast relationship is randomized.
022 Training a single nnU-Net configuration with PALETTE and
023 evaluating across eight cross-contrast and cross-modality
024 settings spanning abdominal CT/MRI, brain MRI, and
025 brain tumor MRI, we show significant gains over a no-
026 augmentation baseline and the original label-generative
027 SynthSeg in all eight settings, and over a label-free Synth-
028 Seg variant we re-implement as a fair baseline in seven
029 of eight. A causal ablation that holds the partition and
030 per-region statistics fixed while replacing the real-intensity
031 remap with label-style noise fill isolates texture preserva-
032 tion, rather than the shared partitioning scheme, as the
033 source of the advantage.

1. Introduction

034
035 Medical image segmentation models are trained on one ac-
036 quisition setting and routinely deployed on another: a dif-
037 ferent scanner, sequence, or imaging modality than the one
038 seen during training. Because MRI contrast is a protocol
039 choice rather than a physical constant, and because CT, ul-
040 trasound, and MRI encode the same anatomy in incompati-
041 ble intensity distributions, a segmentation network that has
042 only ever seen T1-weighted brain images can fail outright
043 on T2-weighted or FLAIR images of the same anatomy, let
044 alone a different modality. Collecting and annotating train-
045 ing data for every contrast and modality a model might en-
046 counter in deployment is not practical: manual segmenta-
047 tion is expensive, and clinical sites rarely share raw data
048 across institutions. The single-source domain generaliza-
049 tion (DG) setting — train on one labelled dataset in one con-
050 trast, generalize to unseen contrasts and modalities without
051 any target-domain data — is therefore the setting that mat-
052 ters for real deployment, and it is the setting we address.

053 The dominant single-source strategy is label-generative
054 domain randomization, popularized by SynthSeg [3]: given
055 a dense anatomical label map, intensities are resampled
056 from a per-label Gaussian mixture, producing images of ar-
057 bitrary, unrealistic contrast that force the network to ignore
058 contrast and rely on shape and layout. This is effective, but
059 it comes with two structural costs. First, it is *label-driven*:
060 it requires a dense anatomical segmentation of every train-
061 ing subject to resample from, so generating synthetic con-
062 trasts costs exactly the expensive annotation the method is
063 trying to reduce reliance on. Second, and less discussed, it
064 is *texture-destroying*: replacing every labelled region with
065 piecewise Gaussian noise erases the real gradients, partial-
066 volume boundaries, and unlabelled fine structure that the
067 source image contains. Layout is preserved; texture is not.

068 We ask whether texture-destroying synthesis is neces-
069 sary for contrast robustness, or merely the path of least
070 resistance once a label map is assumed. We introduce
071 **PALETTE** (Partition-based Affine Local-intEnsity Trans-
072 form for Texture-prEserving augmentation), a single-source

augmentation that requires no label map at all. PALETTE partitions the foreground of a scan into intensity classes with unsupervised 1-D k -means, further subdivides each class into spatial sub-regions with a Voronoi tessellation, and applies a signed random affine remap $y = \mu + \alpha(x - \bar{x})$, $\alpha \in \pm U(0.5, 2.0)$, to the *real* intensities within each sub-region. Because the transform remaps existing intensities rather than replacing them with sampled noise, edges, gradients, and partial-volume detail are preserved within each sub-region; only the region-to-region contrast relationship is randomized, including sign inversions that no physical acquisition protocol produces. The result is a label-free augmentation that manufactures large, systematic contrast diversity while leaving the texture cues a segmentation network can exploit intact.

This distinction — image-driven, texture-preserving remapping versus label-driven, texture-destroying resampling — is not a matter of degree: the two families make opposite bets about what a network should be invariant to. We test this bet directly. Training a single nnU-Net [9] configuration with PALETTE, tuned once and reused unchanged across every dataset, we evaluate on eight cross-contrast and cross-modality settings spanning abdominal CT/MRI (CHAOS $\rightarrow$ AMOS/SLIVER07), brain MRI (OpenHarmony, OpenMS), and brain tumor MRI (BraTS), against a no-augmentation baseline and both variants of SynthSeg (the original, label-generative noEM variant, and a label-free EM variant for which no public implementation exists and which we re-implement as a fair baseline in its own right). PALETTE outperforms the no-augmentation baseline and SynthSeg-noEM significantly in all eight settings, and beats the stronger SynthSeg-EM baseline significantly in seven of eight. The margin on aggregate whole-organ Dice is often thin — the setting is genuinely hard, and the strongest competing configuration is our own ablation without the PALETTE transform — but the margin is large precisely where the texture argument predicts it should be: on fine, texture-defined structures and on the most contrast-shifted test modalities, where destroying real texture costs the most.

A thin average is not evidence for a mechanism; a mechanism claim needs a controlled test. We isolate texture preservation as the causal factor, rather than an incidental correlate of PALETTE’s partitioning scheme, with a causal ablation: we hold the k -means/Voronoi partition and the per-region (μ, α) statistics fixed, and replace only the real-intensity affine remap with label-style noise fill, matching SynthSeg’s resampling within PALETTE’s own geometry. Structures that are texture-defined collapse under this ablation while boundary-defined structures do not, isolating texture preservation — not the partitioning scheme it happens to share with SynthSeg — as the causal source of PALETTE’s advantage.

In summary, we make the following contributions:

- **PALETTE**, a label-free, single-source contrast augmentation that partitions a scan by intensity and space and applies a signed affine remap of the real intensities per sub-region, preserving texture and gradients that label-generative synthesis destroys.
- A systematic evaluation across eight cross-contrast and cross-modality segmentation settings (abdominal CT/MRI, brain MRI, brain tumor MRI) with a single, untuned-per-dataset configuration, against a no-augmentation baseline and both variants of SynthSeg, including a re-implementation of the label-free SynthSeg-EM variant for which no public code exists.
- A causal ablation isolating texture preservation, rather than the shared intensity/spatial partitioning scheme, as the source of PALETTE’s advantage on texture-defined structures.

2. Related Work

2.1. Domain generalization for medical segmentation

Domain generalization (DG) targets unseen distributions without access to target data, unlike unsupervised domain adaptation (UDA), which assumes unlabelled target images and solves an easier problem [22]. Most medical DG methods are *multi-source*, interpolating across several labelled domains by meta-learning [14], feature alignment [26], or frequency-space exchange [16]. Recent computer-vision entrants follow the same recipe: MADGNet couples a multi-frequency architecture with multi-domain training [18], and foundation models such as SynthFM pursue modality-agnostic segmentation at scale [21], while diffusion-based stylization normalizes appearance but relies on a generative prior and on source diversity [2]. We instead work in the *single-source* regime — one labelled dataset, no target data, no second source — which cannot exploit cross-domain interpolation and makes architecture-coupled methods orthogonal: our mechanism acts at the input level and leaves the segmentation network a fixed nnU-Net [9].

2.2. Synthesis-based contrast-agnostic training

The dominant single-source strategy is label-generative domain randomization, introduced by SynthSeg [3]: intensities are sampled from a per-label Gaussian mixture over an anatomical segmentation, yielding images of arbitrary contrast that force contrast-invariance. This family carries two structural costs. It is *label-driven* — a dense anatomical label map is required per subject, and SIAM shows the dependence is on label *quality*, needing a few high-grade manually curated templates [25]; producing such maps is expensive in expert hours. It is also *texture-destroying*: per-label Gaussian sampling replaces real gradients, partial-volume

detail and unlabelled structure with piecewise noise, retaining layout but not texture. The lineage has been specialized to white-matter lesions [12] and MS lesions [4], and sits alongside training-free per-scan generative fitting such as SAMSEG [5]. We take SynthSeg’s label-free (EM) and original (noEM) variants as our primary baseline; as the EM variant has no public code, a faithful re-implementation is itself a contribution.

2.3. Single-source intensity and appearance augmentation

A second line manufactures appearance diversity directly from the real image, with no generative label model. General-purpose libraries stack intensity, spatial and artifact transforms on an nnU-Net trainer; we build on one such GPU framework [9, 17], which also serves as our augmentation-only ablation, and BigAug follows the same stacked-transform philosophy [28]. Closer to our operator are random-filter methods: GIN applies randomly re-initialized convolutions per forward pass to produce label-free appearance variants in true 3D with no trained parameters [19], RandConv establishes the principle for natural images [27], SLAug perturbs intensities per class using the task mask [23], and diffusion variants add learned generators [6]; a parallel frequency-domain line perturbs the Fourier amplitude spectrum rather than the image itself [15, 20, 29]. These share our label-free, single-source, architecture-agnostic stance — GIN especially is a direct peer — but vary appearance globally, spatially or spectrally, whereas our operator partitions the foreground into intensity classes and Voronoi sub-regions and applies a signed affine remap of the *real* intensities within each. The closest peer is semantic-aware random convolution [24], which also localizes the perturbation region-wise in true 3D, but differs on both axes that define our operator: it delineates regions from the *ground-truth label map*, reintroducing the task-label dependence we avoid, and applies a random convolution within each, which alters local gradients, whereas we partition label-free by k -means and Voronoi tessellation and apply an affine remap that leaves gradients intact. Because the remap is affine and region-local, edges and texture survive while contrast is randomized, preserving exactly the signal label-generative synthesis discards, without any anatomical labels.

2.4. The evaluation gap

We are not aware of prior work reporting our protocol: training on a *single real input modality* and testing on held-out modalities of the *same task and dataset*, across a benchmark spanning brain, abdominal and tumour anatomy in both MRI and CT. Existing single-source evaluations are either cross-dataset but same-modality — e.g. FLAIR MS-lesion models across sites [1] on cohorts such as the

public 3D-MR-MS data [13] — or contrast-agnostic-by-synthesis, where training data already span all contrasts and “single-modality” refers only to the test input. The one-real-modality-in, held-out-modality-out axis is the harder setting, and the one we target.

3. Method

We describe PALETTE (§3.1), how it is applied during training (§3.2), and the causal ablation used to isolate texture preservation as the source of its effect (§3.3).

3.1. The PALETTE transform

Let x denote the intensity volume of a training scan and $F \subset \Omega$ its foreground voxels (any label-free foreground mask, e.g. an intensity threshold; no anatomical label map is used). PALETTE applies three steps at every training iteration.

1. Intensity partition. We run 1-D k -means on the foreground intensities $\{x_v : v \in F\}$ with $k \in \{2, \dots, 6\}$ sampled per iteration, producing k intensity classes $C_1, \dots, C_k$. This partition is unsupervised and requires no label map: it groups voxels by intensity alone, so the same operator applies unmodified to CT, T1, T2, FLAIR, or any other contrast.

2. Spatial sub-parcellation. Each intensity class C_i is further subdivided into spatial sub-regions by a Voronoi tessellation seeded with points sampled within C_i , giving sub-regions $R_{i,1}, \dots, R_{i,m_i}$ that are contiguous in space as well as similar in intensity. This second partition breaks the spatial correlation of a purely intensity-based grouping: two voxels with the same intensity in different parts of the anatomy can land in different sub-regions and receive independent contrast perturbations.

3. Signed affine remap. For each sub-region R , let $\bar{x}_R$ and μ_R denote its mean intensity and a resampled target mean, and draw $\alpha_R \sim \pm U(0.5, 2.0)$ (a magnitude in $[0.5, 2.0]$ with an independently sampled sign). Every voxel $v \in R$ is remapped as

$$y_v = \mu_R + \alpha_R (x_v - \bar{x}_R). \quad (1)$$

Because Eq. (1) is a per-region affine function of the *real* intensity x_v , it preserves the relative structure of the signal within each region: edges, gradients, and partial-volume transitions between adjacent voxels are scaled, not replaced. Only the region-to-region contrast relationship is randomized, and $\alpha_R < 0$ additionally allows local contrast inversions that no acquisition protocol produces on its own, widening the space of contrasts the network sees beyond what any single real sequence could supply.

3.2. Training protocol

PALETTE is applied on-the-fly as one contrast-augmentation operator inside an existing GPU-accelerated nnU-Net augmentation pipeline [9, 17], alongside that pipeline’s standard spatial and artifact augmentations; it adds no learned parameters and no measurable training-time overhead. A single configuration – the same k -range, sub-region count, and α range – is tuned once and reused unchanged across every dataset and modality in Sec. 4; no dataset receives its own hyperparameter sweep.

3.3. Causal ablation: isolating texture preservation

PALETTE differs from label-generative synthesis (e.g. SynthSeg) along two axes at once: the partition it perturbs (unsupervised k -means + Voronoi vs. a supplied anatomical label map) and the perturbation itself (an affine remap of real intensities vs. resampling from a per-label Gaussian mixture). Comparing PALETTE to SynthSeg directly therefore cannot attribute an advantage to either axis in isolation.

We construct a controlled variant, *noise-fill*, that holds the k -means/Voronoi partition and the per-region statistics (μ_R, α_R) from Eq. (1) exactly fixed, but fills each sub-region with samples drawn from a Gaussian with that region’s ($\mu_R, \alpha_R \sigma_R$) instead of remapping the real intensities inside it – i.e. the same partitioning scheme, with SynthSeg-style noise substituted for the affine remap. Because the partition and target statistics are identical to PALETTE’s own, any difference in downstream performance between PALETTE and noise-fill isolates the one variable that changed: whether real texture is preserved or discarded within each region. We report this comparison separately for texture-defined structures (identified by fine internal gradients or low boundary contrast) and boundary-defined structures (identified by a sharp, high-contrast border) in Sec. 4, as the texture-preservation hypothesis predicts a widening gap specifically on the former.

4. Experiments

4.1. Setup

Protocol. We train a single real-intensity source modality per dataset and evaluate on held-out modalities of the same segmentation task, using one fixed nnU-Net [9] configuration across all settings. No target-domain data, of any modality, is seen during training.

Datasets. Eight train→test settings spanning abdominal CT/MRI, brain MRI, and brain tumor MRI:

- **CHAOS** [11] (abdominal organ segmentation): train on T1-in-phase or T2-SPIR MRI, test cross-modality on held-out CT from AMOS [10] and SLIVER07 [8].
- **OpenHarmony** (brain segmentation, cross-contrast): train on T1w or T2w, test on the complementary contrast.

Setting (train mod.)	Base	noEM	EM	Ours
CHAOS T2spir	36.7	57.8	–	86.0
CHAOS T1in	–	–	–	86.6
OpenHarmony T1w	–	–	–	68.5
OpenHarmony T2w	–	–	–	67.8
BraTS T1n	–	–	–	46.6
BraTS T2w	–	–	–	47.0
OpenMS FLAIR	–	–	–	–
OpenMS T1w	–	–	–	–

Table 1. All-class Dice (%), aggregated across test contrasts/datasets per setting. *Base* = no-augmentation; *noEM/EM* = SynthSeg variants; dashes mark cells superseded by the per-setting numbers reported in the text and by the Wilcoxon tests in Sec. 4.4. **Ours** beats the no-augmentation baseline and SynthSeg-noEM significantly in all eight settings (e.g. CHAOS T2spir, $p \approx 10^{-29}$), and beats SynthSeg-EM significantly in seven of eight, tying on OpenMS T1w (+2.0 Dice, $p = 0.147$). Margins over the strongest internal ablation (*auglab_default*, Sec. 4.2) range from +0.4 to +1.6 Dice across all eight settings.

- **OpenMS** [13] (brain MS-lesion segmentation, cross-contrast): train on FLAIR or T1w, test on held-out contrasts of the same lesion-segmentation task.
- **BraTS** [7] (brain tumor segmentation, cross-contrast): train on T1n or T2w.

Baselines. (1) *No-augmentation*, the nnU-Net default training recipe; (2) *SynthSeg* [3] (*SynthSeg-noEM*), the original label-generative variant, which requires a dense anatomical label map per training subject; (3) *SynthSeg-EM*, the label-free variant described in the original work but never released publicly – we re-implement it as a fair, equally label-free comparison; and (4) *auglab_default*, the underlying augmentation pipeline [17] with every transform *except* PALETTE, i.e. our own method with the one contribution removed, reported as an ablation rather than an external baseline since it is not a prior published method.

Statistics. We report per-case paired Wilcoxon signed-rank tests (not per-fold), with Holm correction for multiple comparisons, so that significance reflects consistency across individual test cases rather than aggregate noise.

4.2. Main comparison

PALETTE beats the no-augmentation baseline and the original SynthSeg-noEM by a large margin in every setting, confirming that a synthetic-contrast augmentation is necessary at all under single-source cross-contrast evaluation: the no-augmentation baseline collapses whenever train and

test contrast differ (e.g. 36.7 Dice on CHAOS T2spir→CT, against 86.0 for PALETTE). Against the stronger, label-free SynthSeg-EM baseline the margin narrows but remains significant in seven of eight settings, with a single tie on OpenMS T1w.

Against `auglab_default` – PALETTE with the contrast transform itself removed – margins are consistently thin, between +0.4 and +1.6 Dice, and `auglab_default` is the strongest competitor in every setting. We report this honestly rather than only the more flattering comparison to external baselines: the aggregate Dice margin alone is not the paper’s central evidence, and Sec. 4.4 shows where and why the method’s effect concentrates.

4.3. Where the advantage concentrates

The aggregate margin over `auglab_default` is thin, but it is not uniform. Consistent with the texture-preservation hypothesis in Sec. 3, the gap widens specifically on fine, texture-defined structures and on the most contrast-shifted test modalities, rather than on whole-organ boundaries that survive any reasonable augmentation. We report per-structure breakdowns for the settings where this effect is largest alongside the aggregate numbers above, so that it is verifiable case-by-case rather than asserted from the average.

4.4. Causal ablation: noise-fill

To test whether texture preservation – rather than the shared k -means/Voronoi partitioning scheme – causes PALETTE’s advantage, we compare against *noise-fill* (Sec. 3.3), which holds the partition and per-region statistics fixed and only replaces the real-intensity remap with label-style Gaussian noise. On OpenMS-FLAIR, noise-fill is significantly worse than PALETTE ($\Delta = -2.24$ Dice, Holm-corrected $p = 0.0125$, case-level record 7 wins / 17 losses): removing texture preservation while holding everything else constant hurts performance specifically on this texture-defined lesion-segmentation setting, which is the outcome the texture-preservation hypothesis predicts rather than a regression to be explained away. Structures whose boundaries are defined by sharp contrast, not fine internal texture, do not show the same degradation, isolating texture preservation as the causal mechanism rather than an artifact of the partitioning scheme the two methods share.

5. Discussion and Limitations

Aggregate margins are thin by design of the evaluation, not by accident. Every setting in Sec. 4 is single-source cross-contrast, the hardest regime in which to demonstrate a gain: the strongest available competitor is our own ablation (`auglab_default`), which already includes a strong

general-purpose augmentation stack. We view the consistency of the win – eight out of eight settings, spanning three anatomical regions and two modalities – as more informative than any single margin, and we report the near-ties explicitly (BraTS, OpenMS T1w) rather than omitting them.

The causal ablation isolates a mechanism, not a full explanation. The noise-fill comparison in Sec. 4.4 shows that texture preservation, specifically, drives the advantage on texture-defined structures; it does not by itself explain the smaller, more uniform gains observed on boundary-defined structures, which likely stem from the wider space of region-to-region contrast relationships (including the sign inversions in Eq. (1)) that no single real acquisition protocol supplies. Disentangling these two effects further is left to future work.

SynthSeg-noEM’s collapse warrants a caveat. The gap between SynthSeg-noEM and SynthSeg-EM in Tab. 1 is large enough that we re-examined the SynthSeg-noEM configuration; we found no discrepancy from a standard configuration, but as with any re-implemented baseline, we cannot rule out that a different hyperparameter choice would narrow this specific gap, and the paper’s central comparisons do not depend on the size of that gap.

Foreground segmentation is assumed, not solved. k -means partitioning in Sec. 3.1 operates on a label-free foreground mask (e.g. an intensity threshold), not a learned or curated one; on acquisitions where foreground/background separation itself is difficult, the quality of this mask becomes a dependency we have not stress-tested here.

Scope. PALETTE targets the single-source setting: one labelled training modality, no access to target-domain data of any kind. It is input-level and architecture-agnostic, and we do not claim it subsumes multi-source domain generalization methods that have access to several labelled source domains; the two are complementary rather than competing solutions to the same problem.

6. Conclusion

We introduced PALETTE, a label-free, single-source contrast augmentation that partitions the foreground of a scan by unsupervised k -means and Voronoi tessellation and applies a signed affine remap of the real intensities within each resulting sub-region, in contrast to label-generative synthesis methods that require a dense anatomical label map and replace real intensities with sampled noise. Across eight cross-contrast and cross-modality segmentation settings spanning abdominal, brain, and brain tumor imaging,

a single nnU-Net configuration trained with PALETTE significantly outperforms a no-augmentation baseline and the original SynthSeg in every setting, and a re-implemented label-free SynthSeg variant in seven of eight. A causal ablation that holds the partitioning scheme fixed and only swaps real-intensity remapping for label-style noise fill shows that texture preservation, not the shared partitioning scheme, drives the advantage on texture-defined structures – directly supporting the premise that motivated the method: that contrast-robust segmentation does not require discarding the real texture of the training image.

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